MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY TEST – 01** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. If $\cos A = \frac{4}{5}$, then the value of $\tan A$ is: [1]

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $\frac{3}{4}$

Justification: For acute angle A, $\sin A = \sqrt{1-\cos^2 A} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$. Hence $\tan A = \frac{\sin A}{\cos A} = \frac{3/5}{4/5} = \frac{3}{4}$.

Q2. If $\sin\theta = \frac{a}{b}$, then $\cos\theta$ is equal to: [1]

(A) $\frac{b}{\sqrt{b^2-a^2}}$ (B) $\frac{b}{a}$

(C) $\frac{\sqrt{b^2-a^2}}{b}$ (D) $\frac{a}{\sqrt{b^2-a^2}}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $\frac{\sqrt{b^2-a^2}}{b}$

Justification: Using $\sin^2\theta + \cos^2\theta = 1$ (for acute θ), $\cos\theta = \sqrt{1-\frac{a^2}{b^2}} = \frac{\sqrt{b^2-a^2}}{b}$.

Q3. If $\tan\theta = \dfrac{1}{\sqrt{3}}$, then the value of $\dfrac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta}$ is: [1]

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{3}$ (C) $\dfrac{3}{4}$ (D) $\dfrac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\dfrac{1}{2}$

Justification: Here $\tan\theta = \dfrac{1}{\sqrt{3}} \implies \theta = 30°$, so $\csc^2 30° = 4$ and $\sec^2 30° = \dfrac{4}{3}$. Thus $\dfrac{4 - 4/3}{4 + 4/3} = \dfrac{8/3}{16/3} = \dfrac{1}{2}$.

Q4. If $3\cot\theta = 2$, then the value of $\dfrac{4\sin\theta - 3\cos\theta}{2\sin\theta + 6\cos\theta}$ is: [1]

(A) $\dfrac{1}{3}$ (B) $\dfrac{2}{3}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{3}{4}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\dfrac{1}{3}$

Justification: Dividing numerator and denominator by $\sin\theta$ ($\neq 0$): $\dfrac{4 - 3\cot\theta}{2 + 6\cot\theta}$. Substituting $3\cot\theta = 2$ (and $6\cot\theta = 4$) gives $\dfrac{4-2}{2+4} = \dfrac{2}{6} = \dfrac{1}{3}$.

Q5. **Assertion (A):** The value of $\sin\theta = \dfrac{4}{3}$ is not possible for any acute angle θ.

Reason (R): Hypotenuse is the longest side in a right-angled triangle, so $\sin\theta = \dfrac{\text{Perpendicular}}{\text{Hypotenuse}} \leq 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

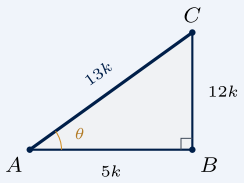
Justification: Since $\dfrac{4}{3} > 1$ and $\sin\theta \leq 1$ in every right-angled triangle, (A) and (R) are both true and (R) is the correct explanation of (A).

Q6. If $\sin\theta = \frac{12}{13}$, evaluate the value of $\frac{2\cos\theta + 3\tan\theta}{\sin\theta}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Right Triangle & Other Ratios [1 Mark]

Let right $\triangle ABC$ have $\angle B = 90^\circ$ and $\angle A = \theta$. Given $\sin\theta = \frac{BC}{AC} = \frac{12}{13}$, take $BC = 12k$ and $AC = 13k$ $(k > 0)$. By Pythagoras' theorem:



$$AB = \sqrt{AC^2 - BC^2} = \sqrt{169k^2 - 144k^2} = 5k$$

Hence $\cos\theta = \frac{5}{13}$ and $\tan\theta = \frac{12}{5}$.

Step 2: Substitution & Simplification [1 Mark]

$$\frac{2\cos\theta + 3\tan\theta}{\sin\theta} = \frac{2\left(\frac{5}{13}\right) + 3\left(\frac{12}{5}\right)}{\frac{12}{13}} = \frac{\frac{10}{13} + \frac{36}{5}}{\frac{12}{13}}$$

$$= \frac{50 + 468}{65} \times \frac{13}{12} = \frac{518}{60} = \frac{259}{30}$$

$$\boxed{\text{Value} = \mathbf{\frac{259}{30}}}$$

Q7. Prove that: $(\csc\theta - \cot\theta)^2 = \frac{1 - \cos\theta}{1 + \cos\theta}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Express LHS in $\sin\theta$ and $\cos\theta$ [1 Mark]

$$\text{LHS} = (\csc\theta - \cot\theta)^2 = \left(\frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta}\right)^2 = \frac{(1 - \cos\theta)^2}{\sin^2\theta}$$

Step 2: Apply $\sin^2\theta = 1 - \cos^2\theta$ & Factorize [1.5 Marks]

$$\text{LHS} = \frac{(1 - \cos\theta)^2}{1 - \cos^2\theta} = \frac{(1 - \cos\theta)^2}{(1 - \cos\theta)(1 + \cos\theta)}$$

Step 3: Cancellation to RHS [0.5 Mark]

$$\boxed{\text{LHS} = \frac{\mathbf{1} - \cos\theta}{\mathbf{1} + \cos\theta} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q8. Prove that: $\dfrac{\cos A}{1+\sin A}+\dfrac{1+\sin A}{\cos A}=2\sec A.$ **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Combine Fractions over Common Denominator [1 Mark]

$$\text{LHS}=\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}=\frac{\cos^2 A+(1+\sin A)^2}{(1+\sin A)\cos A}$$

Step 2: Expand & Simplify Numerator Using $\sin^2 A+\cos^2 A=1$ [1.5 Marks]

$$\text{LHS}=\frac{\cos^2 A+1+2\sin A+\sin^2 A}{(1+\sin A)\cos A}=\frac{1+1+2\sin A}{(1+\sin A)\cos A}=\frac{2(1+\sin A)}{(1+\sin A)\cos A}$$

Step 3: Conclusion [0.5 Mark]

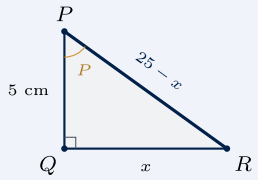
$$\boxed{\text{LHS}=\frac{\mathbf{2}}{\cos \mathbf{A}}=\mathbf{2}\sec \mathbf{A}=\text{RHS}\quad[\text{Hence Proved}]}$$

Q9. In $\triangle PQR$, right-angled at Q, $PR+QR=25$ cm and $PQ=5$ cm. Determine the values of $\sin P$, $\cos P$, and $\tan P$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Apply Pythagoras' Theorem [1.5 Marks]

Let $QR=x$ cm. Then $PR=(25-x)$ cm. In right $\triangle PQR$ ($\angle Q=90^\circ$):



$$PR^2=PQ^2+QR^2 \implies (25-x)^2=5^2+x^2$$

$$625-50x+x^2=25+x^2 \implies 50x=600 \implies x=12 \text{ cm}$$

Thus $QR=12$ cm and $PR=25-12=13$ cm.

Step 2: Calculate Trigonometric Ratios for $\angle P$ [1.5 Marks]

$$\sin P=\frac{QR}{PR}=\frac{12}{13},\quad \cos P=\frac{PQ}{PR}=\frac{5}{13},\quad \tan P=\frac{QR}{PQ}=\frac{12}{5}$$

$$\boxed{\sin P=\frac{\mathbf{12}}{\mathbf{13}},\ \cos P=\frac{\mathbf{5}}{\mathbf{13}},\ \tan P=\frac{\mathbf{12}}{\mathbf{5}}}$$

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Q10. Rooftop Solar Photovoltaic Panel Array Tilt Angle

A solar engineering firm installs rooftop photovoltaic panels. For maximum solar energy absorption at a specific latitude, the panels must be mounted at an angle of $\theta = 30^\circ$ to the horizontal roof. The slant length of each panel module is $AB = 2.4$ m. A vertical strut BC supports the upper edge, while the panel rests on the roof at A, forming a right-angled triangle ABC with $\angle C = 90^\circ$. **Based on the above geometric model, answer the following questions:**

(i) Write the trigonometric ratio connecting vertical height BC, slant length AB, and tilt angle θ. [1]

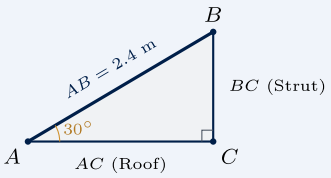
(ii) Calculate the exact height BC of the vertical support strut. [1]

(iii) Find the horizontal roof space AC covered by the panel, and determine the ratio $\dfrac{BC}{AC}$. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Ratio Declaration [1 Mark]

$$\boxed{\sin\theta = \frac{\mathbf{BC}}{\mathbf{AB}}}$$



Sub-part (ii): Vertical Support Height BC [1 Mark]

$$BC = AB\sin 30^\circ = 2.4 \times \frac{1}{2} \implies \boxed{BC = \mathbf{1.2}\text{ m}}$$

Sub-part (iii): Horizontal Roof Space AC & Ratio [2 Marks]

$$AC = AB\cos 30^\circ = 2.4 \times \frac{\sqrt{3}}{2} = \mathbf{1.2\sqrt{3}}\text{ m} \approx 2.08\text{ m}$$

$$\frac{BC}{AC} = \tan 30^\circ \implies \boxed{\text{Ratio } \frac{BC}{AC} = \frac{\mathbf{1}}{\mathbf{\sqrt{3}}}}$$

Note: The diagram caption shows $AB = 2$ m; with $AB = 2$ m, $BC = 1$ m and $AC = \sqrt{3}$ m, while $\frac{BC}{AC} = \frac{1}{\sqrt{3}}$ remains identical.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★